Section A

Answer **all** the questions in this section.

Question 1

Peptic cells from the lining of the mammalian stomach secrete the enzyme precursor pepsinogen. Some of these cells were isolated and maintained in a culture solution containing radioactively labelled amino acids. Samples of the cells were taken at regular intervals and prepared for electron microscopy.

Fig.1.1 shows a drawing from an electron micrograph of a peptic cell. The time taken, in minutes, for radioactivity to be detected in the various cell organelles viewed under the electron microscope is shown on right side of the drawing.

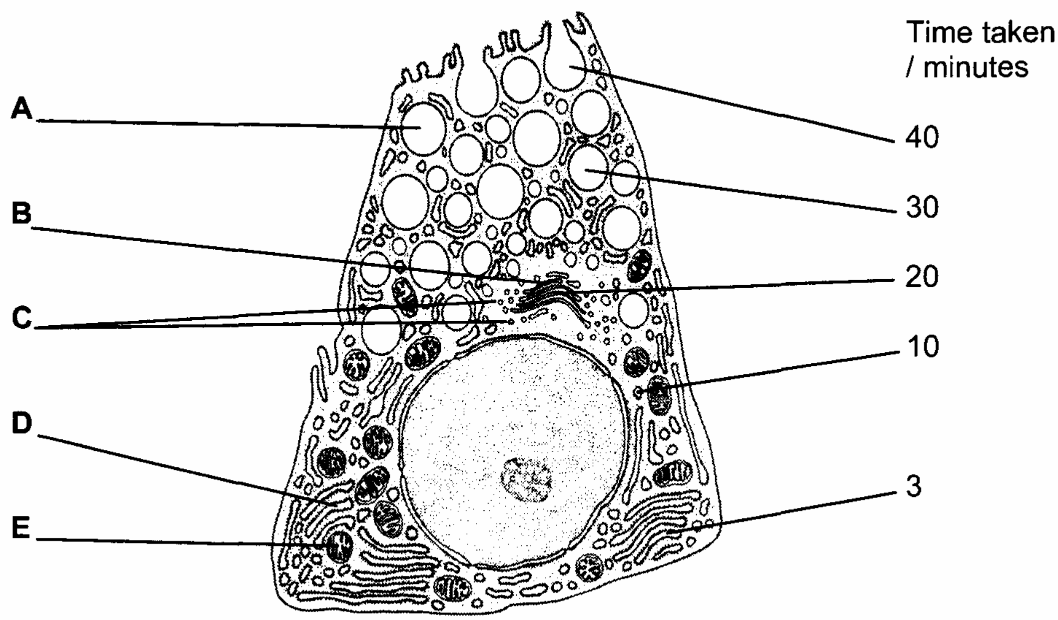

Fig. 1.1

(a) Name the organelles labelled **A** to **D**. [2]

A ....

B ....

C ....

D ....

For Examiner's Use

(b) Explain the different times at which radioactivity are detected in the organelles labelled **B** and **D**. [4]

B ....

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D ....

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(c) Describe how the material in the organelles labelled **A** passes out of the cell. [2]

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(d) Explain why large numbers of the organelle labelled **E** are required in cells synthesising and secreting proteins. [3]

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[Total: 11]

For Examiner's Use

Question 2

Fig. 2.1 and Fig. 2.2 show diagrammatically chromosomes from two cells from the same organism undergoing different types of nuclear division.

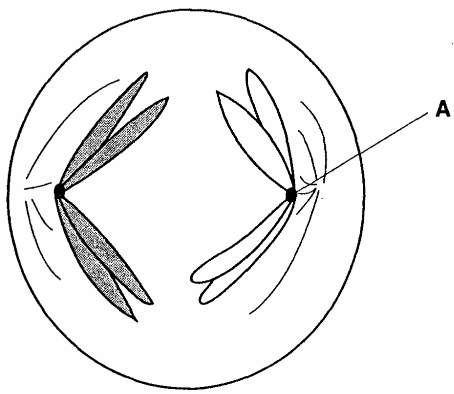

Fig. 2.1

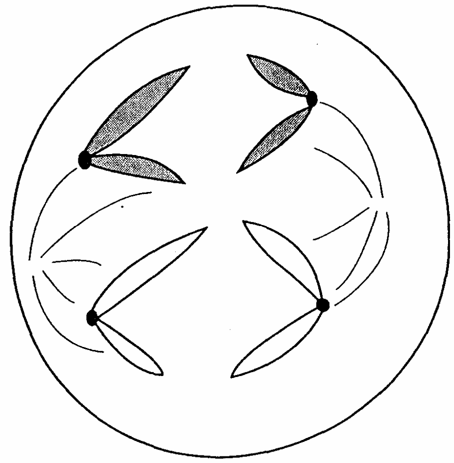

Fig. 2.2

(a)(i) Name structure **A** in Fig. 2.1. [1]

(ii) State **two** functions of structure **A**. [2]

1 --

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2 --

(b) Describe the main difference between the two stages visible in Fig. 2.1 and Fig. 2.2. [2]

For Examiner's Use

(c) Describe what occurs between the stage shown in Fig. 2.2 and cytokinesis (division of cytoplasm). [3]

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Uncontrolled cell division can result in cancer. Some types of cancer can be treated by chemotherapy, which involves the injection of chemicals into the bloodstream.

One chemical used for chemotherapy is called Methotrexate. This is a reversible, competitive inhibitor of one of the enzymes in the metabolic pathway that results in the formation of purines.

(d) Explain how the use of Methotrexate will slow down the mitotic cell cycle. [3]

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[Total: 11]

Question 3

Cystic Fibrosis Transmembrane Regulator (CFTR) is a transmembrane regular protein. Its molecules have 1480 amino acids. People with cystic fibrosis produce defective CFTR protein which is missing one amino acid from its structure. All remaining amino acids remain unchanged.

(a) Which type of gene mutation produced the cystic fibrosis allele? Explain your answer. [2]

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(b) In a particular mutation, one of the bases in the CFTR gene sequence was replaced by another base. However, the CFTR protein remained functional. Suggest **two** reasons how this may be possible. [2]

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2 ....

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Fig. 3.1 shows part of the process of making normal and defective CFTR in a cell. A normal, functional CFTR protein molecule has sugar molecules attached to it.

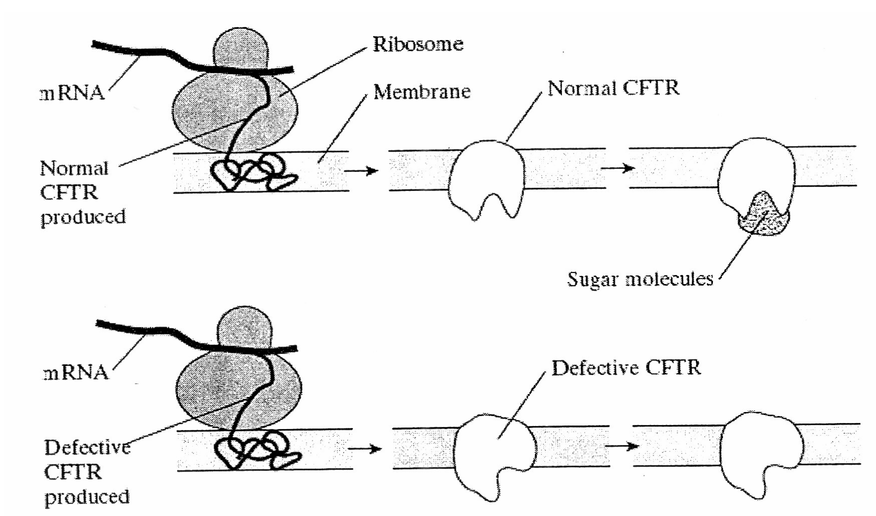

Fig. 3.1

(c) Describe how the information on mRNA is translated into CFTR at the ribosome. [4]

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(d) Using information in Fig. 3.1 and your own knowledge, suggest why defective CFTR, missing one amino acid, is not functional. [2]

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[Total: 10]

For Examiner's Use

Question 4

(a) Describe the term 'operon'. [3]

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Fig. 4.1 shows a pathway that the bacterium, *Escherichia coli* (*E. coli*), uses to metabolize lactose.

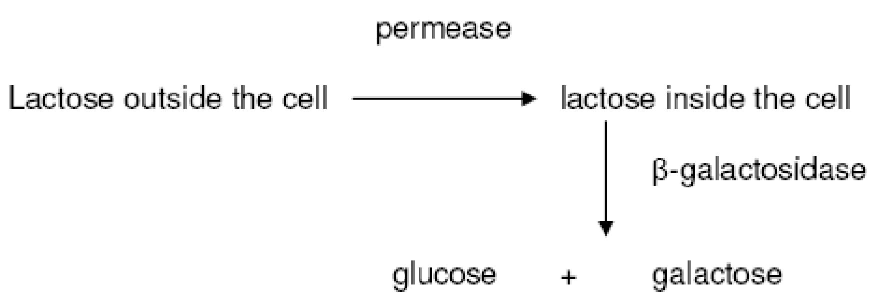

Fig. 4.1

(b) With reference to Fig. 4.1, explain how the pathway is turned 'on' in *E.coli*. [4]

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A student researcher was given a strain of *E. coli* that has its *lac I* gene deleted. Her task was to maximize expression of the gene that codes for β-galactosidase. She has been provided with three types of growth media available to culture her bacteria. Each contains essential salts, vitamins, and other nutrients, but each contains a different selection of sugars as energy sources for the bacteria.

Medium A: Contains fructose, glucose, lactose, maltose, sucrose
Medium B: Contains fructose and glucose
Medium C: Contains lactose

(c) In which medium should she culture her new strain of *E. coli*? Explain your answer. [3]

[Total: 10]

Question 5

Four different eye pigments in *Drosophila* are derived from the amino acid tryptophan. A simplified metabolic pathway of pigment production is shown in Fig. 5.1. Three different gene loci, **V**/**v**, **C**/**c** and **B**/**b** are involved in this pathway, acting as shown in Fig. 5.1.

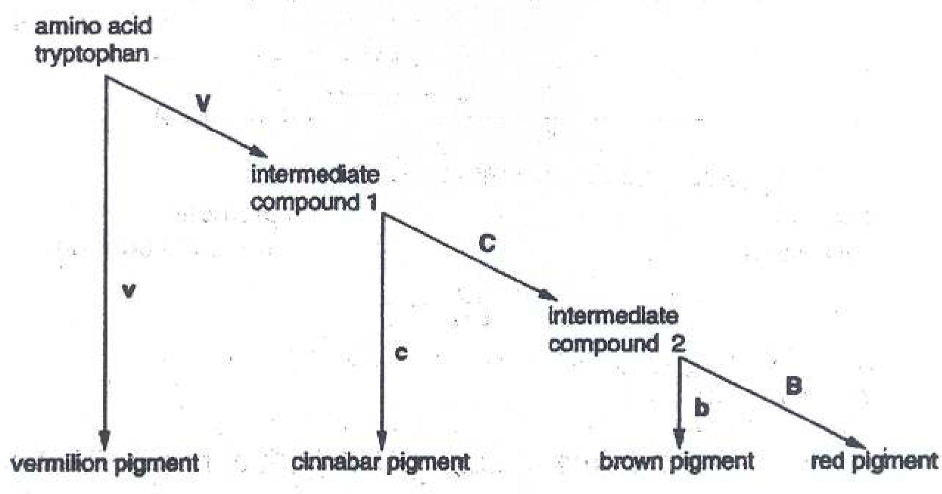

Fig. 5.1

(a)(i) Using the information in Fig. 5.1, deduce the phenotypes of flies with the following genotypes: [2]

genotype	*phenotype*
VvCcBb	
vvCCBB	

(ii) State the term that is applied to this type of gene interaction. [1]

(iii) Suggest an explanation for the gene interaction in the pathway of pigment production. [3]

Pure-breeding red-eyed flies, **CCBB** were crossed with pure-breeding cinnabar-eyed flies, **ccbb**. (Both sets of flies were homozygous, **VV**, so this locus can be ignored.) The F1 were test crossed giving red-eyed, brown-eyed and cinnabar-eyed flies.

(b)(i) Draw a genetic diagram in the space below to show the expected genotypes and phenotypes of the offspring of this test-cross, and state the ratio of phenotypes expected. Assume that the gene loci are **not** linked. (Take care that the symbols **C** and **c** cannot be confused in your answer.) [5]

In fact, the **C**/**c** and **B**/**b** loci are both on chromosome 2 of *Drosophila*, although not very close together.

(ii) Explain how this linkage might affect the expected results of the test-cross. [3]

[Total: 14]

Question 6

Fig 6.1 shows the relationship between various metabolic processes.

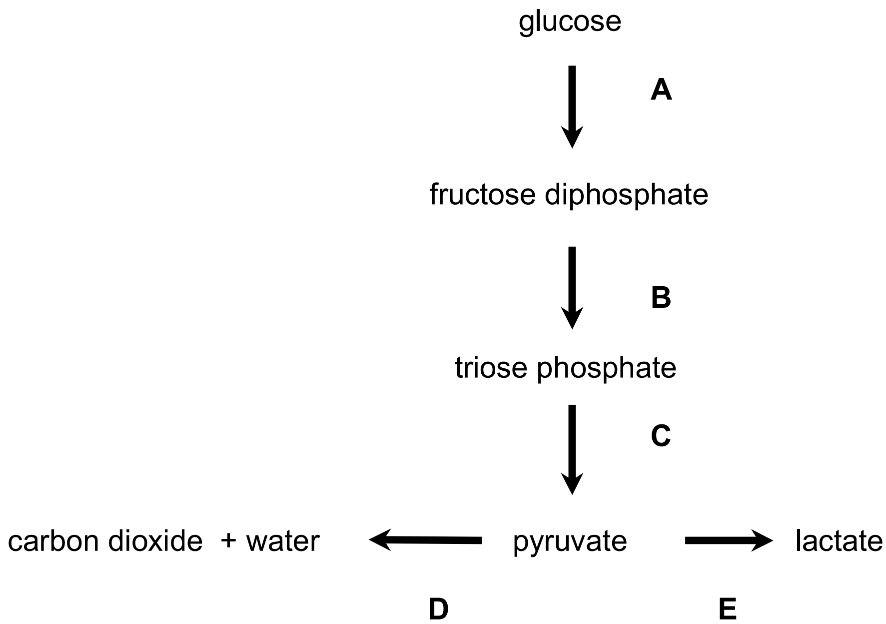

Fig. 6.1

(a) At which stage(s) would NAD be reduced? [1]

(b) Explain the significance of pathway **E**. [2]

In an investigation, mammalian liver cells were homogenized and the resulting homogenate centrifuged. Portions containing only nuclei, ribosomes, mitochondria and cytosol were each isolated. Samples of each portion, and of the complete homogenate, were incubated in four ways.

1. with glucose
2. with pyruvate
3. with glucose plus cyanide
4. with pyruvate plus cyanide

After incubation, the presence or absence of carbon dioxide and lactate in each sample was determined. The results are summarised in Table 6.1.

Table 6.1

	samples of homogenate									
	complete		nuclei only		ribosome only		mitochondria only		cytosol	
	CO_2	lactate	CO_2	lactate	CO_2	lactate	CO_2	lactate	CO_2	lactate
glucose	✓	✓	X	X	X	X	X	X	X	✓
pyruvate	✓	✓	X	X	X	X	✓	X	X	✓
glucose + cyanide	X	✓	X	X	X	X	X	X	X	✓
pyruvate + cyanide	X	✓	X	X	X	X	X	X	X	✓

(c)(i) With reference to this investigation, name **two** organelles not involved in respiration. [2]

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(ii) Explain why carbon dioxide is produced when mitochondria are incubated with pyruvate but not when incubated with glucose. [3]

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(iii) Explain why, in the presence of cyanide, lactate production does occur, but not carbon dioxide production. [3]

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For Examiner's Use

(d) Describe and explain **one** physical way in which the inner membrane of a mitochondrion is adapted to assist in the production of ATP. [3]

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[Total: 14]

Question 7

The peppered moth, *Biston betularia*, is found in Britain. Figs. 7.1, 7.2 and 7.3 show the photographs of three varieties of this moth.

Fig. 7.1 (*carbonaria*)

Fig. 7.2 (*insularia*)

Fig. 7.3 (*typica*)

The colour of the moth is determined by a gene which has three alleles. The allele **CC** results in the *carbonaria* variety and is dominant to the other two alleles. The allele $\mathbf{C^i}$ causes the *insularia* variety. $\mathbf{C^i}$ is dominant to the allele $\mathbf{C^t}$ which causes the *typica* variety.

(a) Explain how crosses between *insularia* moths could produce some *insularia* and some *typica* offspring, but no *carbonaria* offspring; [2]

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A study was carried out in the middle of the 20th century. The frequencies of two of the varieties of the peppered moth were investigated in two cities in Britain, rural Dorset and in the industrial city of Birmingham. The results were as follows.

	Frequency of variety	
	carbonaria	***typica***
Dorset	0.000	0.946
Birmingham	0.850	0.102

At that time, many lichens on the bark of trees in Birmingham had been killed by industrial pollution. The tree bark and other surfaces were often blackened by soot. The area studied in Dorset was free of industrial pollution. Peppered moths often rest on exposed surfaces during the daytime.

(b) Explain how the process of natural selection might have led to the different frequencies of the varieties of the peppered moth seen in Dorset and Birmingham. [4]

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The three types of peppered moth are classified as varieties of the same species.

(c)(i) How is it possible to show that the three varieties belong to the same species? [2]

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For Examiner's Use

(ii) Over long periods of time, it is possible for varieties of one species to develop into separate species. Suggest how this could happen in moths such as the peppered moth. [2]

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[Total: 10]

Section B

Answer **ONE** question.

Write your answers on the separate writing papers provided.
Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.
Your answers must be in continuous prose, where appropriate.
Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

Question 8

(a) Describe control elements and explain how they influence transcription. [7]

(b) Contrast between prokaryotic DNA and eukaryotic chromosome. [7]

(c) Explain how changes at the genome level can lead to cancer. [6]

[Total: 20]

Question 9

(a) Discuss roles played by proteins in maintaining the potential differences across membranes and in the transmission of nerve impulses along an axon. [7]

(b) Explain how an action potential may be generated in a postsynaptic membrane. [7]

(c) Describe the series of reactions that take place after a glucagon hormone binds to its target cell causing the overall increase of blood sugar level. [6]

[Total: 20]

End of H2 Paper 2